

Exponential One-Sided Deviations for the Stake-Volume Chain

A self-contained solution note

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Abstract

Tang and Yao's paper on polynomial voting rules leaves open the problem of proving the expected exponential one-sided deviation bounds for the total stake-volume Markov chain in the interval not covered by their path-counting estimates. This note proves those bounds. More precisely, for the chain

$$N_{t+1} = N_t + \mathbf{1}\{\text{success at time } t\}, \quad \mathbb{P}(\text{success at time } t \mid N_t) = N_t^{-\alpha},$$

with $\alpha > 0$, let

$$\beta := \frac{1}{1+\alpha}, \quad \kappa := (1+\alpha)^{1/(1+\alpha)}.$$

We prove that, at speed t^β ,

$$\mathbb{P}\{N_t \leq \lambda t^\beta\} \leq \exp\{-c_-(\lambda)t^\beta + o(t^\beta)\} \quad \text{for every } 0 < \lambda < \kappa,$$

and

$$\mathbb{P}\{N_t \geq \lambda t^\beta\} \leq \exp\{-c_+(\lambda)t^\beta + o(t^\beta)\} \quad \text{for every } \lambda > \kappa,$$

where $c_-(\lambda), c_+(\lambda) > 0$ are given by explicit variational formulae. Thus the result is a solution of the natural one-sided exponential-deviation gap in Tang–Yao's open problem. It is not a full large-deviation principle: matching lower bounds and an exact rate function are not proved here. The proof is based on the hitting-time representation of N_t as a sum of independent geometric waiting times.

1 Status of the result

The contribution of this note is a *solution*, not merely partial progress, for the following natural interpretation of the Tang–Yao question: prove exponential upper bounds for the lower tail below the true fluid threshold and for the upper tail above the true fluid threshold. The true threshold is

$$\kappa = (1+\alpha)^{1/(1+\alpha)}.$$

Thus the meaningful statements are

$$\mathbb{P}\{N_t \leq \lambda t^{1/(1+\alpha)}\} \leq e^{-ct^{1/(1+\alpha)}} \quad (0 < \lambda < \kappa),$$

and

$$\mathbb{P}\{N_t \geq \lambda t^{1/(1+\alpha)}\} \leq e^{-ct^{1/(1+\alpha)}} \quad (\lambda > \kappa),$$

for suitable $c > 0$.

The result should not be advertised as a complete large-deviation principle. It gives sharp-threshold one-sided exponential upper bounds at the expected speed, but it does not prove matching lower bounds. Also, if the open problem were read literally as asking for both one-sided exponential upper bounds for every λ in an interval containing κ , that literal formulation would be false: one of the two events has probability tending to one on each side of κ .

2 Model and main theorem

Fix $\alpha > 0$ and an initial value $N_0 = N \in \mathbb{N}_+$. Consider the discrete-time nondecreasing Markov chain

$$N_{t+1} = \begin{cases} N_t + 1, & \text{with probability } N_t^{-\alpha}, \\ N_t, & \text{with probability } 1 - N_t^{-\alpha}, \end{cases} \quad t = 0, 1, 2, \dots \quad (1)$$

Set

$$\beta := \frac{1}{1 + \alpha}, \quad \kappa := (1 + \alpha)^\beta. \quad (2)$$

The heuristic differential equation $dn/dt = n^{-\alpha}$ gives

$$n(t) \sim ((1 + \alpha)t)^\beta = \kappa t^\beta,$$

so κ is the expected threshold between the lower and upper tails.

For $0 < s < 1$, define

$$A_\alpha(s) := \int_0^1 -\log(1 - sx^\alpha) dx. \quad (3)$$

For $s > 0$, define

$$B_\alpha(s) := \int_0^1 \log(1 + sx^\alpha) dx. \quad (4)$$

For $0 < \lambda < \kappa$, put

$$I_-(\lambda) := \sup_{0 < s < 1} \left\{ \frac{s}{\lambda^{1+\alpha}} - A_\alpha(s) \right\}, \quad c_-(\lambda) := \lambda I_-(\lambda). \quad (5)$$

For $\lambda > \kappa$, put

$$I_+(\lambda) := \sup_{s > 0} \left\{ B_\alpha(s) - \frac{s}{\lambda^{1+\alpha}} \right\}, \quad c_+(\lambda) := \lambda I_+(\lambda). \quad (6)$$

Theorem 2.1 (Exponential one-sided deviation bounds). *For the Markov chain (1), the following hold.*

(i) *If $0 < \lambda < \kappa$, then $c_-(\lambda) > 0$ and*

$$\limsup_{t \rightarrow \infty} t^{-\beta} \log \mathbb{P}\{N_t \leq \lambda t^\beta\} \leq -c_-(\lambda) < 0. \quad (7)$$

Equivalently, for every $0 < c < c_-(\lambda)$ there exists t_0 such that, for all $t \geq t_0$,

$$\mathbb{P}\{N_t \leq \lambda t^\beta\} \leq e^{-ct^\beta}.$$

(ii) *If $\lambda > \kappa$, then $c_+(\lambda) > 0$ and*

$$\limsup_{t \rightarrow \infty} t^{-\beta} \log \mathbb{P}\{N_t \geq \lambda t^\beta\} \leq -c_+(\lambda) < 0. \quad (8)$$

Equivalently, for every $0 < c < c_+(\lambda)$ there exists t_0 such that, for all $t \geq t_0$,

$$\mathbb{P}\{N_t \geq \lambda t^\beta\} \leq e^{-ct^\beta}.$$

Remark 2.2 (Optimizers and explicit constants). *The constants in Theorem 2.1 are explicit. Since*

$$A'_\alpha(s) = \int_0^1 \frac{x^\alpha}{1 - sx^\alpha} dx, \quad 0 < s < 1,$$

and A'_α is continuous and strictly increasing from $(1 + \alpha)^{-1}$ to $+\infty$, the supremum in (5) is attained at the unique point $s_-(\lambda) \in (0, 1)$ satisfying

$$A'_\alpha(s_-(\lambda)) = \lambda^{-(1+\alpha)}.$$

Similarly,

$$B'_\alpha(s) = \int_0^1 \frac{x^\alpha}{1 + sx^\alpha} dx, \quad s > 0,$$

and B'_α is continuous and strictly decreasing from $(1 + \alpha)^{-1}$ to 0. Hence the supremum in (6) is attained at the unique point $s_+(\lambda) > 0$ satisfying

$$B'_\alpha(s_+(\lambda)) = \lambda^{-(1+\alpha)}.$$

3 Hitting-time representation

We use geometric random variables supported on $\{1, 2, \dots\}$. Let $(G_n)_{n \geq N}$ be independent random variables with

$$G_n \sim \text{Geom}(n^{-\alpha}), \quad \mathbb{P}(G_n = k) = n^{-\alpha}(1 - n^{-\alpha})^{k-1}, \quad k = 1, 2, \dots$$

When the chain is at state n , the waiting time for the next increase has exactly the distribution of G_n . By the strong Markov property these successive waiting times are independent. Define, for $m > N$,

$$T_m := \sum_{n=N}^{m-1} G_n, \tag{9}$$

and set $T_N := 0$. Then T_m is the first time at which the chain reaches state m . Consequently,

$$\{N_t \geq m\} = \{T_m \leq t\}, \quad \{N_t \leq m - 1\} = \{T_m > t\}. \tag{10}$$

Thus the problem becomes a one-sided large-deviation problem for sums of independent, non-identically distributed geometric variables.

For $p_n = n^{-\alpha}$, the moment generating functions are

$$\mathbb{E}e^{\theta G_n} = \frac{p_n e^\theta}{1 - (1 - p_n)e^\theta}, \quad \theta < -\log(1 - p_n), \tag{11}$$

$$\mathbb{E}e^{-\theta G_n} = \frac{p_n e^{-\theta}}{1 - (1 - p_n)e^{-\theta}}, \quad \theta > 0. \tag{12}$$

The natural exponential tilting scale is $\theta = sm^{-\alpha}$ when $m \asymp t^\beta$, because then $m^{-\alpha}t \asymp m$.

4 Log-Laplace estimates

Lemma 4.1 (Scaled log-Laplace estimates). *Fix $\alpha > 0$ and the initial state N .*

(i) *For every $0 < s < 1$,*

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \sum_{n=N}^{m-1} \log \mathbb{E} \exp\{sm^{-\alpha} G_n\} \leq A_\alpha(s). \quad (13)$$

(ii) *For every $s > 0$,*

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \sum_{n=N}^{m-1} \log \mathbb{E} \exp\{-sm^{-\alpha} G_n\} \leq -B_\alpha(s). \quad (14)$$

Proof. Write $x_{n,m} := n/m$.

First consider the positive transform. Let $\theta_m = sm^{-\alpha}$, with $0 < s < 1$. If $n_m/m \rightarrow x \in (0, 1]$, then (11) gives

$$\begin{aligned} \mathbb{E} e^{\theta_m G_{n_m}} &= \frac{n_m^{-\alpha} e^{sm^{-\alpha}}}{1 - (1 - n_m^{-\alpha}) e^{sm^{-\alpha}}} \\ &\rightarrow \frac{x^{-\alpha}}{x^{-\alpha} - s} = \frac{1}{1 - sx^\alpha}. \end{aligned}$$

This convergence is uniform for $x \in [\varepsilon, 1]$, for each fixed $\varepsilon > 0$. Hence the contribution of the indices with $n/m \in [\varepsilon, 1]$ converges to

$$\int_{\varepsilon}^1 -\log(1 - sx^\alpha) dx.$$

It remains to control the indices with $n/m < \varepsilon$. Choose s' such that $s < s' < 1$. For all sufficiently large m ,

$$e^{sm^{-\alpha}} - 1 \leq s' m^{-\alpha}.$$

Therefore

$$\begin{aligned} 1 - (1 - n^{-\alpha}) e^{\theta_m} &= n^{-\alpha} - (1 - n^{-\alpha})(e^{\theta_m} - 1) \\ &\geq n^{-\alpha} - s' m^{-\alpha} = n^{-\alpha} (1 - s' x_{n,m}^\alpha). \end{aligned}$$

Using (11), we obtain

$$\mathbb{E} e^{\theta_m G_n} \leq \frac{e^{\theta_m}}{1 - s' x_{n,m}^\alpha}. \quad (15)$$

For $n/m < \varepsilon$, the logarithm of the right-hand side is at most

$$\theta_m - \log(1 - s' \varepsilon^\alpha),$$

provided ε is small enough that $s' \varepsilon^\alpha < 1$. After summing over at most $\varepsilon m + O(1)$ such indices and dividing by m , the resulting upper bound is

$$\varepsilon [-\log(1 - s' \varepsilon^\alpha)] + o(1),$$

which tends to 0 as $\varepsilon \downarrow 0$. Combining the large- x convergence with this small- x bound proves (13).

For the negative transform, again put $\theta_m = sm^{-\alpha}$, now with $s > 0$. If $n_m/m \rightarrow x \in (0, 1]$, then (12) gives

$$\begin{aligned}\mathbb{E}e^{-\theta_m G_{n_m}} &= \frac{n_m^{-\alpha} e^{-sm^{-\alpha}}}{1 - (1 - n_m^{-\alpha})e^{-sm^{-\alpha}}} \\ &\longrightarrow \frac{x^{-\alpha}}{x^{-\alpha} + s} = \frac{1}{1 + sx^\alpha},\end{aligned}$$

again uniformly for $x \in [\varepsilon, 1]$. For the small- x part we use only the trivial inequality

$$\mathbb{E}e^{-\theta_m G_n} \leq 1,$$

so those logarithmic terms are at most zero. Consequently, for every fixed $\varepsilon > 0$,

$$\limsup_{m \rightarrow \infty} \frac{1}{m} \sum_{n=N}^{m-1} \log \mathbb{E}e^{-sm^{-\alpha} G_n} \leq \int_{\varepsilon}^1 -\log(1 + sx^\alpha) dx.$$

Letting $\varepsilon \downarrow 0$ proves (14). □

5 Proof of the main theorem

Proof of Theorem 2.1. We first prove the lower-tail estimate. Fix $0 < \lambda < \kappa$ and define

$$m_t := \lfloor \lambda t^\beta \rfloor + 1.$$

For all sufficiently large t , $m_t > N$. Since N_t is integer-valued,

$$\{N_t \leq \lambda t^\beta\} \subseteq \{N_t \leq m_t - 1\} = \{T_{m_t} > t\}.$$

For any fixed $0 < s < 1$, Chernoff's inequality with $\theta_t = sm_t^{-\alpha}$ gives

$$\mathbb{P}(T_{m_t} > t) \leq e^{-\theta_t t} \prod_{n=N}^{m_t-1} \mathbb{E}e^{\theta_t G_n}.$$

Because $m_t/t^\beta \rightarrow \lambda$ and $\beta(1 + \alpha) = 1$,

$$\frac{t}{m_t^{1+\alpha}} \longrightarrow \lambda^{-(1+\alpha)}.$$

Dividing by m_t and applying Lemma 4.1(i), we obtain

$$\limsup_{t \rightarrow \infty} \frac{1}{m_t} \log \mathbb{P}(T_{m_t} > t) \leq -\frac{s}{\lambda^{1+\alpha}} + A_\alpha(s).$$

Multiplying by $m_t/t^\beta \rightarrow \lambda$ yields

$$\limsup_{t \rightarrow \infty} t^{-\beta} \log \mathbb{P}\{N_t \leq \lambda t^\beta\} \leq -\lambda \left\{ \frac{s}{\lambda^{1+\alpha}} - A_\alpha(s) \right\}. \quad (16)$$

Taking the infimum of the right-hand side over $0 < s < 1$ proves (7).

It remains to check that the exponent is strictly negative. As $s \downarrow 0$,

$$A_\alpha(s) = \int_0^1 (sx^\alpha + O(s^2x^{2\alpha})) dx = \frac{s}{1+\alpha} + O(s^2).$$

The condition $\lambda < \kappa$ is equivalent to

$$\lambda^{-(1+\alpha)} > \frac{1}{1+\alpha}.$$

Hence, for all sufficiently small positive s ,

$$\frac{s}{\lambda^{1+\alpha}} - A_\alpha(s) > 0.$$

Thus $I_-(\lambda) > 0$ and $c_-(\lambda) > 0$.

We now prove the upper-tail estimate. Fix $\lambda > \kappa$ and define

$$m_t := \lceil \lambda t^\beta \rceil.$$

Again $m_t > N$ for all sufficiently large t , and since N_t is integer-valued,

$$\{N_t \geq \lambda t^\beta\} \subseteq \{N_t \geq m_t\} = \{T_{m_t} \leq t\}.$$

For any fixed $s > 0$, Chernoff's inequality with $\theta_t = sm_t^{-\alpha}$ gives

$$\mathbb{P}(T_{m_t} \leq t) \leq e^{\theta_t t} \prod_{n=N}^{m_t-1} \mathbb{E} e^{-\theta_t G_n}.$$

Using $t/m_t^{1+\alpha} \rightarrow \lambda^{-(1+\alpha)}$ and Lemma 4.1(ii),

$$\limsup_{t \rightarrow \infty} \frac{1}{m_t} \log \mathbb{P}(T_{m_t} \leq t) \leq \frac{s}{\lambda^{1+\alpha}} - B_\alpha(s).$$

Therefore

$$\limsup_{t \rightarrow \infty} t^{-\beta} \log \mathbb{P}\{N_t \geq \lambda t^\beta\} \leq -\lambda \left\{ B_\alpha(s) - \frac{s}{\lambda^{1+\alpha}} \right\}. \quad (17)$$

Optimizing over $s > 0$ proves (8).

Finally,

$$B_\alpha(s) = \int_0^1 (sx^\alpha + O(s^2x^{2\alpha})) dx = \frac{s}{1+\alpha} + O(s^2), \quad s \downarrow 0.$$

The condition $\lambda > \kappa$ is equivalent to

$$\lambda^{-(1+\alpha)} < \frac{1}{1+\alpha}.$$

Thus, for all sufficiently small positive s ,

$$B_\alpha(s) - \frac{s}{\lambda^{1+\alpha}} > 0,$$

and consequently $I_+(\lambda) > 0$ and $c_+(\lambda) > 0$. □

6 Almost sure fluid limit

The preceding exponential bounds immediately imply the almost sure law of large numbers at the correct scale.

Corollary 6.1 (Almost sure fluid limit). *For every initial state $N \in \mathbb{N}_+$,*

$$\frac{N_t}{t^\beta} \longrightarrow \kappa \quad \text{almost surely.}$$

Proof. Fix rational numbers q, r with $0 < q < \kappa < r$. By Theorem 2.1, there are constants $c_q, c_r > 0$ and t_0 such that, for all $t \geq t_0$,

$$\mathbb{P}\{N_t \leq qt^\beta\} \leq e^{-c_q t^\beta}, \quad \mathbb{P}\{N_t \geq rt^\beta\} \leq e^{-c_r t^\beta}.$$

Since $\beta > 0$, both right-hand sides are summable over integer t . Borel–Cantelli implies that, almost surely, for all sufficiently large t ,

$$q < \frac{N_t}{t^\beta} < r.$$

Taking the intersection over the countable set of rational pairs $q < \kappa < r$ gives

$$\liminf_{t \rightarrow \infty} \frac{N_t}{t^\beta} \geq \kappa, \quad \limsup_{t \rightarrow \infty} \frac{N_t}{t^\beta} \leq \kappa,$$

and hence the asserted limit. □

7 Why a literal two-sided reading is impossible

The bounds proved above are the only possible exponential upper bounds of this type. A statement asking for both

$$\mathbb{P}\{N_t \leq \lambda t^\beta\} \leq e^{-ct^\beta} \quad \text{and} \quad \mathbb{P}\{N_t \geq \lambda t^\beta\} \leq e^{-ct^\beta}$$

with $c > 0$ for every λ in an interval containing κ cannot be correct.

Indeed, Corollary 6.1 implies that if $\lambda > \kappa$, then

$$\mathbb{P}\{N_t \leq \lambda t^\beta\} \longrightarrow 1,$$

so no positive exponential upper bound can hold for that lower-tail event. Similarly, if $\lambda < \kappa$, then

$$\mathbb{P}\{N_t \geq \lambda t^\beta\} \longrightarrow 1,$$

so no positive exponential upper bound can hold for that upper-tail event. At $\lambda = \kappa$ itself, one should not expect a strictly positive t^β -speed exponent for either side.

8 Relation with the Tang–Yao interval

Tang and Yao’s path-counting estimates involve the function

$$f_\alpha(\lambda) := (1 + \alpha)\lambda \log \lambda - (1 + \alpha)\lambda + \lambda^{-\alpha}. \tag{18}$$

Let $\lambda_-(\alpha) < \lambda_+(\alpha)$ denote its two positive roots. We verify that the threshold $\kappa = (1 + \alpha)^{1/(1+\alpha)}$ lies strictly between those roots.

Proposition 8.1. *For every $\alpha > 0$,*

$$\lambda_-(\alpha) < \kappa < \lambda_+(\alpha).$$

Consequently, Theorem 2.1 gives exponential bounds throughout the missing pieces

$$\lambda \in (\lambda_-(\alpha), \kappa) \quad \text{for the lower tail,}$$

and

$$\lambda \in (\kappa, \lambda_+(\alpha)) \quad \text{for the upper tail.}$$

Proof. First,

$$\frac{f_\alpha(\kappa)}{\kappa} = (1 + \alpha) \log \kappa - (1 + \alpha) + \kappa^{-(1+\alpha)}.$$

Using $(1 + \alpha) \log \kappa = \log(1 + \alpha)$ and $\kappa^{-(1+\alpha)} = (1 + \alpha)^{-1}$, we get

$$\frac{f_\alpha(\kappa)}{\kappa} = \log(1 + \alpha) - (1 + \alpha) + \frac{1}{1 + \alpha}.$$

The function $x \mapsto \log x - x + x^{-1}$ is zero at $x = 1$ and has derivative

$$\frac{1}{x} - 1 - \frac{1}{x^2} < 0 \quad (x > 1),$$

so $f_\alpha(\kappa) < 0$ for $\alpha > 0$. Also,

$$f_\alpha(\lambda) \rightarrow +\infty \quad \text{as } \lambda \downarrow 0, \quad f_\alpha(\lambda) \rightarrow +\infty \quad \text{as } \lambda \rightarrow \infty.$$

Finally,

$$f_\alpha''(\lambda) = \frac{1 + \alpha}{\lambda} + \alpha(1 + \alpha)\lambda^{-\alpha-2} > 0, \quad \lambda > 0.$$

Thus f_α is strictly convex, is negative at κ , and tends to $+\infty$ at both ends of $(0, \infty)$. Its two positive roots therefore straddle κ . $\square$

9 The case $\alpha = 0$

If $\alpha = 0$, the transition probability is identically one. Hence $N_t = N + t$ deterministically, $\beta = 1$, and $\kappa = 1$. For every $\lambda < 1$,

$$\mathbb{P}\{N_t \leq \lambda t\} = 0$$

for all sufficiently large t , and for every $\lambda > 1$,

$$\mathbb{P}\{N_t \geq \lambda t\} = 0$$

for all sufficiently large t . Thus the desired one-sided exponential upper bounds are trivial in the degenerate case $\alpha = 0$.

References

- [1] Wenpin Tang and David D. Yao. Polynomial Voting Rules. *Mathematics of Operations Research* 50(1):90–106, 2025. Published online February 7, 2024. DOI: [10.1287/moor.2023.0080](https://doi.org/10.1287/moor.2023.0080).